Synopsis: node2vec — Scalable Feature Learning for Networks

Reference: Grover, A., & Leskovec, J. (2016). *node2vec: Scalable Feature Learning for Networks.* Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.

1. Question

Tasks like node classification and link prediction depend on good feature representations of nodes and edges. Most existing approaches rely on hand-engineered features or on fixed ways of exploring a network, so they fail to capture the different kinds of connectivity patterns that exist. Two of the most relevant are homophily and structural equivalence. Homophily means that nodes that are connected or close to one another tend to share similar characteristics, so their representations should be similar, as often happens within a community. Structural equivalence means that nodes playing similar roles in the network should look alike, even when they belong to different communities.

The paper asks how to learn flexible, low-dimensional node representations that capture both properties. The authors' answer is node2vec, a scalable framework that explores neighbourhoods through biased random walks.

2. Method

Node2vec treats the network as a graph $G = (V, E)$ and maps every node into a low-dimensional feature space. The goal is to maximise the likelihood of preserving the sampled network neighbourhoods, using a Skip-gram-inspired objective optimised with stochastic gradient descent (SGD).

The core idea is a biased second-order random walk with two parameters. The return parameter p sets how likely the walk is to step back to the node it just came from, while the in-out parameter q controls the likelihood of exploring nodes farther from the previous node, relative to remaining close to the previous node. Tuning them lets the walk shift between breadth-first search (BFS)-like behaviour, which captures structural equivalence, and depth-first search (DFS)-like behaviour, which captures homophily and community structure.

Two assumptions underpin the framework: neighbourhood nodes are conditionally independent given the source node's representation, and source and neighbour nodes influence each other symmetrically in feature space. Negative sampling keeps the optimisation efficient. For link prediction, the node representations can be merged with binary operators to form edge representations.

3. Key Result / Evidence

The authors tested node2vec on real-world networks, including BlogCatalog, a human protein-protein interaction (PPI) network, and a Wikipedia word co-occurrence network. They compared multi-label classification and link prediction performance against DeepWalk, LINE, and spectral clustering, among others.

In multi-label classification, with half of the nodes labelled for training, node2vec reached Macro-F1 scores of 0.2581 on BlogCatalog, 0.1791 on PPI, and 0.1552 on Wikipedia. DeepWalk was the closest competitor, and node2vec achieved competitive or superior performance across the reported classification experiments, though its margin over DeepWalk on PPI was small at 1.3%. For link prediction on the arXiv collaboration network, it improved AUC by up to 12.6% over the best heuristic baseline, Adamic-Adar. The experiments also showed that the method holds up when edges are missing or noisy, and that using more and longer random walks generally improves results.

4. Limitation and Why It Matters

Node2vec has a few limitations. Reusing random-walk samples saves computation but can bias the sampling. Choosing good values for p and q may need labelled data and extra computation. The authors also admit they don't yet know why the Hadamard operator works so well for combining edge features, and they leave that for future work.

These limitations matter because they affect how far the results can be trusted and repeated. When labels are scarce, tuning p and q becomes difficult, so the method may not reach its best performance in exactly the settings where it is needed most, and biased sampling plus extra tuning make results harder to reproduce. Overall, node2vec is a flexible and scalable approach to network representation learning, but how well it performs depends on a sensible exploration strategy and careful experimental setup.